Proposed Mission to the First Exoplanet in Space-Focusing on Shishen-4

The goal of this mission is to find an exoplanet most likely to provide a safe habitat, scientific benefits, and the lowest degree of risk (in terms of human life) traveling through the vast distance of interstellar space. In assessing the seven candidate planets, it is clear that Shishen-4 is the most promising and logical choice for this pioneering journey.

Parent Star Environment and Orbital Habitability

It is extremely important that the parent star be stable and spectrally similar to our sun to ensure a sustained habitat for generations to come. Shishen-4 is found to be orbiting a G-type star, exactly like our sun. G-type stars have an extremely long estimated lifetime, on the order of 8.9-17.5 billion years (Gyr). Additionally, it has an orbital distance of 1.1 Astronomical Units (AU), which falls well within the generally accepted habitable zone for a G-type star, defined as ranging from 0.9-1.8 AU. Shishen-4 is located conveniently within the life-sustaining region, thus avoiding the more extreme temperature conditions of the inner or outer edges.

Planet Type and Physical Characteristics

Shishen-4's physical attributes are what truly makes it a prime target for human occupation. It possesses a mass 1.15 times that of Earth ($1.15m_E$) and a radius 1.08 times that of Earth ($1.08r_E$), giving it a density of $5.03g/cm^3$. Compared to densities of rocky planets in our own solar system (Mercury: $5.4g/cm^3$, Earth: $5.52g/cm^3$, Mars: $3.9g/cm^3$), Shishen-4's is very close, signifying it is a rocky planet and not an ice or gas giant. As it is rocky and has only slightly more mass than Earth, it qualifies as a "super-earth".

Surface Water

An exoplanet's ability to retain liquid surface water hinges on its gravitational pull (in order to hold a significant atmosphere) and its proximity to its sun. Shishen-4, being a super-earth located well within the habitable zone of its parent star, meets both criteria. Unlike Lockyer-2, which is considered a sub-earth (0.107mE), the larger planet, Shishen-4, possesses the necessary gravitational strength to adequately retain an atmosphere and water; on the other hand, Lockyer-2's orbital distance (1.5AU) likely leads to the majority of its water remaining in frozen form.

Age and Evolutionary State

Exobiology of an exoplanet cannot truly be studied without considering its age, which can tell us about how evolved its biosphere (or life) is likely to be. Shishen-4 is estimated to be 1.9Gyr old. Based on how life is believed to evolve on Earth, a planet at this age should be in the Archaean geological era and contain exclusively single-celled life forms. This is a considerable advantage for mission planning, unlike the age of Tusi-1250 (8 Gyr) or Cannon-01 (8.6 Gyr), which are in the Astrozoic eon and likely possess super-terrestrial intelligence that would pose an unacceptably high risk to the crew.

Distance from Earth and Crew Consideration

Finally, when accounting for the immense distances and the practical aspects of human space travel, Shishen-4's distance solidifies it as the superior target. At only 11 light-years (Ly) away and traveling at 20% the speed of light (0.20c), the trip to Shishen-4 would take exactly 55 years, in stark contrast to Maya-186's distance of 38 Ly (which would be a 190-year journey).

The crew of the first interstellar ship can be comforted knowing that this journey falls within the lifespan of a human. Although sophisticated life support systems, potentially cryogenic stasis, or even a two-generation mission architecture are required, the psychological ease afforded by knowing that the destination can be reached within decades rather than centuries cannot be underestimated. Shishen-4 is, without a doubt, the safest, most feasible, and most compelling mission destination for humanity.